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ARTICLE

Investigation of Factors Influencing Electric Vehicle Adoption in Indonesia: EV Owners' Perspectives

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Abstract

Electric vehicle (EV) uptake in Indonesia remains markedly below policy benchmarks. This study applies the Unified Theory of Acceptance and Use of Technology version 3 (UTAUT3), an extension of UTAUT2 that incorporates personal innovativeness as an additional construct to examine its impact on both behavioral intention and actual EV adoption within the Indonesian context. Unlike studies that typically survey the general public, this study focuses on actual EV users and owners, providing more representative and responsive insights into real-world EV usage. A total of 208 respondents participated, with 135 from the Jabodetabek area and 73 from Surabaya. The UTAUT3 framework explains 60.7% of the variance in behavioral intention and 62.5% in actual EV adoption in Indonesia. The results indicate that facilitating conditions, habit, personal innovativeness, and behavioral intention, significantly influence EV use behavior. Behavioral intention is positively influenced by hedonic motivation, price value, performance expectancy, and personal innovativeness, while social influence exerts a significant negative effect. Although facilitating conditions and habit do not significantly affect intention, they directly impact adoption behavior. These findings provide strategic implications for industry actors and policymakers aiming to accelerate EV adoption in the Indonesian market.

Keywords: Electric vehicle, Behavioral intention, Use behavior, Indonesia

1. Introduction

The Indonesian government has introduced electric vehicle (EV) programs to reduce transportation-related emissions, with an ambitious target of deploying 2 million electric cars by 2030. As of mid-2024, however, EV penetration remains far below this target, with fewer than 30,000 EVs registered nationwide despite a 104% increase in sales compared to the previous year.[1] This disparity between policy ambitions and real-world adoption highlights the challenges of meeting national goals.

Despite the government's initiatives, public acceptance remains a critical determinant of adoption. Previous research shows that consumer perceptions and attitudes strongly shape the adoption process (Shalender & Sharma, 2021; Vafaei-Zadeh et al., 2022). Understanding the factors influencing Indonesians' interest in EVs is essential. Previous research highlights multiple factors affecting EV adoption, including range, incentives, reliability, affordability, environmental benefits, vehicle performance, infrastructure, and cost (Hamzah et al., 2022; Higueras-Castillo et al., 2021; Jain et al., 2022;

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Zarazua de Rubens, 2019). Some studies also examine psychological aspects such as attitudes toward behavior, moral norms, environmental concerns, perceived behavioral control, and subjective norms. Increasingly, research emphasizes the importance of consumer attitudes and behaviors in EV adoption (Lee et al., 2023; Shalender & Sharma, 2021; Shanmugavel & Balakrishnan, 2023; Vafaei-Zadeh et al., 2022).

Indonesia poses distinctive challenges for EV adoption. The nation's vast geographic dispersion, limited and unevenly distributed charging infrastructure, heavy reliance on fossil fuels, and relatively lower purchasing power compared to developed economies constitute significant structural barriers to widespread EV uptake. These socio-economic and infrastructural conditions position Indonesia as a critical case study for examining EV adoption in emerging markets, where national ambitions to deploy 2 million electric cars and 13 million electric motorbikes by 2030 as part of decarbonization efforts must be reconciled with localized constraints.

Previous research on EV adoption has largely focused on advanced economies such as Europe, North America, and Australia, where infrastructure and regulatory frameworks are more mature. These studies have analyzed consumer preferences among different groups, including the general public, new internal combustion engine (ICE) vehicle buyers, EV testers, and EV owners (Axsen et al., 2017; Azarova et al., 2020; Bailey et al., 2015; Globisch et al., 2019). Each group contributes unique insights depending on their degree of involvement. Studies involving the general public or ICE buyers suggest that these individuals have limited knowledge and engagement with EVs. In contrast, those who have tested EVs can provide insights into aspects like comfort, performance, and technology, but their experiences do not fully represent real-world usage (Jensen et al., 2013). EV owners, however, offer the most relevant insights into actual travel behaviors, infrastructure needs, and charging station availability (Axsen et al., 2017; Pan et al., 2019).

Distinct from prior studies, this research investigates the determinants of EV adoption within the context of a developing country, offering insights that may differ from those observed in more advanced markets. Challenges and opportunities vary significantly between nations. Developed countries typically have well-established infrastructure, stricter emissions regulations, and greater incentives for EV adoption. In contrast, developing countries face infrastructure limitations and lower purchasing power, hindering adoption. Policymakers in these regions are still in the process of formulating strategies to encourage EV uptake. This research stands out by surveying EV owners and users while also analyzing differences between developing nations and developed nations. It aims to generate context-specific insights to inform marketing strategies, accelerate EV adoption, and guide policy formulation.

Centered on the emerging market of Indonesia, the research examines key factors driving EV adoption by synthesizing insights from prior literature and empirical feedback from current users. The findings are anticipated to identify critical determinants influencing both EV uptake. These insights are essential for policymakers and industry stakeholders in designing effective frameworks to support EV adoption and expand the national EV market share.

2. Literature review

Environmental, economic, social, and technological factors influence the adoption of EVs. These factors can serve as either drivers or barriers to EV adoption. Environmentally conscious consumers are more likely to adopt EVs, as evidenced by previous research (Cui et al., 2021; He et al., 2018; Patil & Majumdar, 2022). In parallel, economic factors substantially influence adoption decisions. Prospective users tend to assess various cost-related components, including the initial purchase price, charging fees, maintenance expenditures, and battery replacement costs, prior to making a commitment (Giansoldati et al., 2020; Utami et al., 2020; Zhu et al., 2019). Empirical findings further indicate that lowering the overall cost of EV ownership significantly enhances the likelihood of adoption (Lin & Sovacool, 2020; Noel et al., 2020). Social factors influence consumer behavior regarding EV adoption. These include emotional experiences related to product use and the association of EV ownership with self-identity or social status (Han et al., 2017; Nilsson & Nykvist, 2016; İmre et al., 2022).

Technological considerations, such as the advancement of charging infrastructure, vehicle design, driver assistance systems, battery performance, operational efficiency, safety, maintenance requirements, and range capabilities, are also critical in facilitating the transition to EVs (Jain et al., 2022; Patil & Majumdar, 2022; Siddique et al., 2022; Wu et al., 2021; Zhang et al., 2022; İmre et al., 2022). Numerous studies have examined the interrelationships among these technological attributes to better understand the key drivers influencing technology acceptance and utilization. Interrelated factor models have been extensively employed as analytical frameworks for formulating strategies to accelerate EV adoption.

Studies on EV adoption have employed both qualitative and quantitative approaches, with the latter being more prevalent. Researchers have applied various models to examine influencing factors, as illustrated in Table 1. Among the most widely acknowledged frameworks are the Technology Acceptance Model (TAM), the Theory of Planned Behavior (TPB), the Unified Technology Acceptance Model (UTAM), and the Unified Theory of Acceptance and Use of Technology (UTAUT), along with its extended versions (UTAUT2 and UTAUT3). Some studies have also incorporated additional variables outside these frameworks, tailoring them to specific research objectives.

Before presenting the comparison of previous studies (Table 1), the variables commonly used in adoption research are clarified as follows: UB (Use Behavior), BI (Behavioral Intention), SI (Social Influence), PV (Price Value), PI (Personal Innovativeness), PE (Performance Expectancy), HM (Hedonic Motivation), H (Habit), FC (Facilitating Conditions), and EE (Effort Expectancy).

This study advances the literature by applying the UTAUT3 model in a developing-country context. UTAUT3 extends UTAUT2 by incorporating personal innovativeness, reflecting individuals' willingness to explore and adopt emerging technologies. This addition is particularly relevant for EVs, which remain relatively novel in Indonesia.

Drawing upon the UTAUT3 framework, this study posits twelve paths reflecting the relationships among key constructs influencing EV adoption:

H1= Performance expectancy has a positive effect on the intention to adopt EVs.
H2 = Effort expectancy positively affects the intention to adopt EVs.
H3= Social influence significantly and positively impacts the intention to adopt EVs.
H4= Facilitating conditions positively influence the intention to adopt EVs.
H5= Facilitating conditions have a direct positive effect on actual EV usage behavior.
H6= Hedonic motivation exerts a positive influence on the intention to adopt EVs.
H7= Price value significantly enhances the intention to adopt EVs.
H8= Habit has a positive effect on the intention to adopt EVs.
H9= Habit positively influences actual EV usage behavior.
H10= Personal innovativeness contributes positively to the intention to adopt EVs.
H11= Personal innovativeness also positively affects actual usage behavior.
H12= Behavioral intention positively influences actual usage behavior of EVs.

This study advances the existing body of knowledge by extending the application of the UTAUT3 model to a developing country context, specifically Indonesia, an underrepresented region in EV adoption research. Unlike prior studies such as Zhou et al. (2021) that integrated personal innovativeness but focused on mature EV markets, this research gathers empirical data from actual EV users in the Jabodetabek (Jakarta, Bogor, Depok, Tangerang, Bekasi) and Surabaya metropolitan areas. By capturing insights from experienced adopters in an emerging market, the study offers more practical, context-specific, and policy-relevant findings. These insights are particularly valuable for informing national strategies, improving public infrastructure planning, and accelerating EV market penetration in Indonesia and similar developing economies.

3. Materials and methods

3.1. Samples collection

Data were collected via a questionnaire, which took approximately 13 min to complete. The questionnaire was distributed both digitally and physically between June and September 2024. Online distribution was conducted through EV communities on social media, while offline distribution involved direct engagement with EV users at car exhibitions and public charging stations. Participants were electric car users and owners aged 17 years or older who possessed a valid driver's license. They resided in two major metropolitan areas of Indonesia, Surabaya and Jabodetabek, which have the highest concentration of EV users in the country. This facilitated an adequate sample size that accurately represents the situation in Indonesia. Respondents were selected through purposive sampling, as the absence of a national or provincial database of EV users in Indonesia made random sampling unfeasible.

Content validity was ensured through a preliminary pilot survey. The survey comprises two primary elements: The initial section requested participants to furnish socio-demographic data,

Table 1. Insights from previous studies.

Author (year)	Theoretical model	Country	Number of respondents	Respondent				Analytical method	Variables used									
				Electric vehicle users and owners	People are aware of electric vehicles	Electric vehicle driver	Potential user		UB	BI	SI	PV	PI	PE	HM	H	FC	EE
Morton et al. (2016)	Diffusion of innovation theory	United Kingdom	400		√			CB-SEM and logistic regression			√							
He et al. (2018)	–	Cina	369		√			PLS		√			√					
Zhou et al. (2019)	UTAUT	Cina	366			√		SEM	√	√	√			√			√	√
Lee et al. (2021)	–	Pakistan	359		√			PLS-SEM		√	√			√			√	√
Khazaei and Tareq (2021)	UTAUT2	Malaysia	322		√		√	SEM-BSEM		√	√		√				√	
Zhou et al. (2021)	UTAUT2	Cina	725			√		SEM	√	√	√	√		√	√	√	√	√
Abbasi et al. (2021)	UTAUT	Malaysia	199				√	PLS-SEM		√	√			√				√
Jain et al. (2022)	UTAUT	India	284	√			√	CB-SEM and OLS regression		√	√			√			√	√
Shanmugavel and Micheal (2022)	TAM	India	402				√	SEM					√					
Vafaei-Zadeh et al. (2022)	C-TAM-TPB	Malaysia	213		√			PLS-SEM		√		√						
Manutworakit and Choocharukul (2022)	UTAUT	Thailand	403		√			PLS-SEM	√	√	√	√		√	√		√	√
Singh et al. (2023)	UTAUT2	India	309				√	SEM		√	√	√		√	√	√	√	√
Higueras-Castillo et al. (2023)	UTAUT2-VBN	India & Spanyol	643		√			SEM		√	√			√	√		√	√
Habib et al. (2024)		Arab Saudi	397	√			√	PLS SEM		√	√			√				
García de Blanes Sebastián et al., 2024	Meta-UTAUT	Spanyol	326	√				SEM	√		√			√			√	√

encompassing gender, age, residence, education, occupation, marital status, monthly expenditures, quantity of ICE vehicles, number of EVs owned, frequency of EV utilization, and duration of EV ownership. The second section assessed perceptions related to the determinants of EV adoption, following the UTAUT3 framework (Farooq et al., 2017). Respondents were asked to rate their agreement with a series of statements using a five-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree). The questionnaire items for each construct were adapted from validated instruments used in prior empirical studies, as detailed in Table 2.

3.2. Data analysis

Data analysis was carried out using SmartPLS, a widely adopted tool for Partial Least Squares Structural Equation Modeling (PLS-SEM), particularly in social science research contexts (Hair et al., 2019). The PLS-SEM analysis included validity assessment and hypothesis testing. Data processing was performed using Smart PLS v3.2.6, which enabled the detection of complex relationships between constructs and their indicators.

Before conducting hypothesis testing, the reliability and validity of all constructs were assessed. A construct was considered reliable and valid if it met the following criteria: Cronbach's alpha, factor loadings, and Composite Reliability (CR) values exceeding 0.70. Convergent validity was evaluated using the Average Variance Extracted (AVE), with values above 0.50 indicating an acceptable level. The structural model's explanatory power was assessed through the coefficient of determination (R^2), where values of 0.25 or higher were deemed substantial, as recommended by Hair et al. (2014). To ensure discriminant validity, the Heterotrait-Monotrait (HTMT) ratio was employed; HTMT values below 0.90 confirmed adequate discriminant validity, following the threshold suggested by Gold et al. (2001).

4. Results and discussion

4.1. Descriptive statistics

This research employs a total sample of 208 EV owners and users. The sample size is considered adequate for survey-based research and SEM analysis (Hibberts et al., 2012; Memon et al., 2020). Of the total sample, 135 respondents are from Jabodetabek, while 73 are from Surabaya. Table 3 provides a comprehensive overview of the respondents' socio-demographic profiles.

Most EV owners and users in this study are male. The majority of respondents are aged 30−39, hold a bachelor's degree, and own at least one ICE vehicle. Married individuals constitute the primary group of EV owners. Respondents reported using their EVs for work-related travel, school drop-offs, and pickups. Most respondents own a single EV that has been in operation for less than one year. These results indicate that the diffusion of electric vehicles in Indonesia remains at an early development phase.

4.2. Specific responses on electric vehicle usage

Prior to testing the proposed hypotheses, both convergent and discriminant validity were assessed. Construct reliability was established through indicator loadings, all of which exceeded the 0.7 benchmark. Convergent validity was supported by AVE values ranging from 0.551 to 0.859, indicating that each latent construct accounts for more than 50% of the variance in its respective indicators. Composite reliability values, which fell between 0.729 and 0.950, further confirmed the internal consistency of the constructs. A comprehensive summary of indicator loadings, AVE scores, and composite reliability is provided in Table 4. The complete HTMT values are shown in Table 5.

4.3. Structural model

To evaluate construct consistency, VIF values were calculated, all of which were below the threshold of 3.3, suggesting no multicollinearity concerns. The structural model was assessed through the coefficient of determination (R^2) and the significance of path coefficients. R^2 values indicate the proportion of variance in the dependent constructs explained by the model, typically categorized as substantial ($\geq$0.75), moderate ($\geq$0.50), or weak ($\geq$0.25). In evaluating the explanatory power of the model, the R^2 values for behavioral intention (0.607) and use behavior (0.625) exceeded the generally accepted minimum threshold of 0.25 for substantial explanatory power in behavioral research (Hair et al., 2016). These values indicate that the model provides a robust explanation of variance in the target constructs, supporting the reliability of the findings. Specifically, behavioral intention (0.607) was influenced by performance expectancy, effort expectancy, facilitating conditions, social influence, price value, hedonic motivation, habit, and personal innovativeness. Conversely, use behavior (0.625) was

Table 2. Description of constructs and item sources.

Item ID	Item	Sources
Behavior intention		
BI1	I am likely to continue using electric vehicles as my primary mode of transportation.	(Venkatesh et al., 2012; Zhou et al., 2021)
BI2	I plan to frequently use electric vehicles for my daily activities.	
Effort expectancy		
EE1	Electric vehicles are simple to operate	(Jain et al., 2022; Venkatesh et al., 2012; Zhou et al., 2021)
EE2	The functions and features of electric vehicles are easy for me to understand.	
EE3	It is simple to learn how to operate electric vehicles	
EE4	It is simple to become proficient in using electric vehicles	
Facilitating conditions		
FC1	There is a 24-h contact service in case of issues	(Jain et al., 2022; Khazaei & Tareq, 2021; Venkatesh et al., 2012; Zhou et al., 2021)
FC2	It is simple to buy and sell electric vehicles	
FC3	Availability of public electric vehicle charging stations	
FC4	Electric vehicles are well-integrated with other technologies I regularly use, such as smartphones.	
FC5	Technical support, mechanics, and spare parts are readily available	
Habit		
H1	I am accustomed to keeping up with the types and developments of vehicles, including EVs	(Venkatesh et al., 2012; Zhou et al., 2021)
H2	I tend to develop habits of switching between different types of vehicles.	
H3	Using electric vehicles has become a habit	
H4	I have a hobby of collecting different types of vehicles (including electric cars)	
H5	I quickly get addicted to new products, including electric vehicles	
Hedonic motivation		
HM1	Using electric vehicles brings a sense of pride	(Khazaei & Tareq, 2021; Venkatesh et al., 2012)
HM2	My use of electric vehicles reflects my identity	
HM3	I find using electric vehicles to be a pleasant and enjoyable experience.	
HM4	Using electric vehicles is very exciting	
Performance expectancy		
PE1	Electric vehicles can improve work productivity	(Venkatesh et al., 2012; Zhou et al., 2021)
PE2	Electric vehicles are helpful in daily activities	
Personal innovativeness		
PI1	I am generally enthusiastic about trying out new technologies.	(Dutta et al., 2015; Khazaei & Tareq, 2021)
PI2	I am curious about and willing to try the advanced features offered by electric vehicles.	
PI3	I enjoy experimenting with the sophistication of electric vehicle technology	
PI4	I like comparing new technology with existing technology	
Price value		
PV1	Government subsidies make electric vehicles more affordable.	(Venkatesh et al., 2012; Wang et al., 2018; Zhou et al., 2021)
PV2	The purchase price for electric vehicles fits my budget	
PV3	Electric vehicles have reasonable maintenance costs	
PV4	I perceive the price of electric vehicles as fair in relation to the benefits they provide.	
Social influence		
SI1	The use of electric vehicles follows modern societal trends	(Venkatesh et al., 2012; Wang et al., 2018; *Zhou* et al., 2021)
SI2	People around me support and encourage the adoption of electric vehicles.	
SI3	Promotional content through advertisements and social media influences my interest in electric vehicles.	
SI4	Government recommendations or programs influence the use of electric vehicles	
SI5	Environments where many people already use EVs impact the use of electric vehicles	
Use Behavior		
UB1	I use electric vehicles to support government programs	Jain et al., 2022; Khazaei & Tareq, 2021
UB2	I use electric vehicles to reduce environmental pollution	
UB3	I use electric vehicles to facilitate city travel	
UB4	I use electric vehicles for long-distance travel	

Table 3. Descriptive data characteristics.

Description	Frequency (people)	Percentage (%)
Gender		
Male	184	88
Female	24	12
Age		
17–29 years	60	29
30–39 years	66	32
40–49 years	55	26
≥50 years	27	13
Marital status		
Married	148	71
Unmarried	60	29
Education		
≤ High School	34	16
Bachelor's degree	119	57
Master's degree	49	24
Doctoral degree	6	3
Monthly expenditure, IDR		
≤10 million	54	26
10.1 million - 20 million	81	39
20.1 million - 30 million	33	16
30.1 million - 40 million	17	8
>40 million	23	11
ICE vehicles owned		
0	16	8
1	117	56
2	48	23
3	18	9
>3	9	4
EVs owned		
1	197	94.7
2	10	4.8
3	1	0.5
EV driving experience		
≤1 year	112	54
1–2 years	78	38
2–3 years	15	7
>3 years	3	1

Table 4. Measurement model.

Item ID	Factor Loading	CR	AVE
BI1	0.931	0.838	0.859
BI2	0.922		
EE1	0.816	0.876	0.704
EE2	0.759		
EE3	0.888		
EE4	0.886		
FC1	0.778	0.862	0.616
FC2	0.814		
FC3	0.742		
FC4	0.785		
FC5	0.804		
H1	0.786	0.844	0.582
H2	0.765		
H3	0.728		
H4	0.754		
H5	0.782		
HM1	0.705	0.900	0.635
HM2	0.711		
HM3	0.858		
HM4	0.896		
PE1	0.862	0.807	0.802
PE2	0.928		
PI1	0.810	0.864	0.703
PI2	0.881		
PI3	0.834		
PI4	0.828		
PV1	0.712	0.762	0.576
PV2	0.756		
PV3	0.788		
PV4	0.778		
SI1	0.829	0.950	0.703
SI2	0.856		
SI3	0.837		
SI4	0.848		
SI5	0.821		
UB1	0.773	0.728	0.551
UB2	0.781		
UB3	0.704		
UB4	0.705		

predicted by facilitating conditions, habit, personal innovativeness, and behavioral intention.

Hypotheses were evaluated based on the estimated path coefficients, p-values, and t-values derived from the bootstrapping procedure with 5000 resamples at a 5% significance level. A hypothesis was considered statistically significant if the t-value exceeded the critical value of 1.97 and the p-value was below 0.05. According to Hair et al. (2014), a path coefficient greater than 0.1 indicates a meaningful relationship between constructs within the model. The analysis results revealed that nine out of twelve hypotheses were supported. Specifically, behavioral intention, facilitating conditions, and habit exhibited significant positive effects on electric vehicle use behavior, whereas personal innovativeness did not show a statistically significant influence. Furthermore, behavioral intention was significantly predicted by performance expectancy, effort expectancy, hedonic motivation, price value, social influence, and personal innovativeness. In contrast, habit and facilitating conditions did not significantly influence behavioral intention. A detailed summary of the hypothesis testing results is provided in Table 6.

To explore the influence of demographic factors on adoption intention, a Multi-Group Analysis (MGA) was conducted. MGA facilitates the comparison of structural model relationships across distinct subgroups within a single study. In this research, the demographic variables examined included age, marital status, education level, monthly expenditure, and prior electric vehicle driving experience. The analysis revealed that age significantly moderated the relationships between facilitating conditions and behavioral intention,

Table 5. Discriminant validity using HTMT ratio.

	BI	EE	FC	H	HM	PE	PI	PV	SI	UB
BI										
EE	0.667									
FC	0.411	0.471								
H	0.337	0.359	0.860							
HM	0.653	0.612	0.713	0.782						
PE	0.747	0.668	0.595	0.463	0.653					
PI	0.586	0.490	0.651	0.808	0.818	0.518				
PV	0.803	0.659	0.774	0.741	0.787	0.790	0.731			
SI	0.189	0.306	0.706	0.855	0.669	0.423	0.591	0.528		
UB	0.811	0.700	0.745	0.772	0.831	0.644	0.826	0.855	0.721	

Table 6. Hypothesis results on electric vehicle adoption.

Hyphotesis	Path	Hypothesis test		T statistics	VIF	Result
		Coefficients	P values			
H1	BI → UB	0.397	0.000	5.565	1.367	*Supported*
H2	EE → BI	0.155	0.055	1.921	1.746	*Not Supported*
H3	FC → BI	−0.090	0.348	0.939	2.541	*Not Supported*
H4	FC → UB	0.180	0.019	2.341	2.192	*Supported*
H5	H → BI	−0.152	0.142	1.469	3.666	*Not Supported*
H6	H → UB	0.216	0.020	2.324	2.922	*Supported*
H7	HM → BI	0.252	0.006	2.764	2.689	*Supported*
H8	PE → BI	0.249	0.000	3.577	1.855	*Supported*
H9	PI → BI	0.205	0.004	2.845	2.442	*Supported*
H10	PI → UB	0.200	0.039	2.062	2.381	*Supported*
H11	PV → BI	0.355	0.002	3.090	2.563	*Supported*
H12	SI → BI	−0.175	0.015	2.426	2.008	*Supported*

habit and use behavior, as well as personal innovativeness and use behavior. Additionally, monthly expenditure was found to significantly moderate the relationship between personal innovativeness and use behavior. EV driving experience was found to significantly moderate the relationship between habit and behavioral intention. In contrast, marital status and education level did not demonstrate significant moderating effects on any of the structural paths. Detailed results regarding the moderating effects of these control variables are presented in Table 7.

4.4. Discussion

This study employs the UTAUT3 model to examine the determinants of electric vehicle usage, revealing that habit, facilitating conditions, personal innovativeness, and behavioral intention significantly influence actual usage behavior. Among these factors, behavioral intention emerges as the primary predictor of electric car adoption among Indonesian consumers. Furthermore, behavioral intention is positively and significantly influenced by performance expectancy, price value, hedonic motivation, and personal innovativeness, while social influence exerts a significant but negative effect.

Although this research surveyed EV owners rather than non-users, their experiences provide indirect insights into the broader public acceptance issue. Respondents reported skepticism from peers and communities, particularly regarding affordability, charging infrastructure, and long-term reliability. Such perspectives reveal that while early adopters embrace EVs, public acceptance remains a key barrier to mainstream adoption.

Table 6 presents the significance levels of these constructs, offering insights into their relative influence and explanatory power. Notably, behavioral intention exhibits a coefficient of 0.397 (p-value <0.05), underscoring its strong and meaningful impact on EV purchase decisions. This intention is positively affected by variables including performance expectancy, price value, hedonic motivation, and personal innovativeness, while social influence exerts a significant negative effect. Behavioral intention to adopt electric vehicles (EVs) is shaped by multiple determinants, with performance expectancy emerging as a primary influencing factor. The analysis indicates that performance expectancy has a coefficient of 0.249 (p-value <0.05), reflecting a statistically significant relationship with the intention to purchase an EV. In this context, performance expectancy pertains to the anticipated benefits of

Table 7. Moderating effects of control variables.

Demographic variable		BI→UB	EE→BI	FC→BI	FC→UB	H→BI	H→UB	HM→BI	PE→BI	PI→BI	PI→UB	PV→BI	SI→BI
Age	≤40 years	0.000	0.080	0.775	0.002	0.11	0.001	0.012	0.018	0.038	0.847	0.271	0.246
	>40 years	0	0.204	0.019	0.036	0.381	0.446	0.182	0.001	0.185	0.002	0.029	0.121
	PLS-MGA	0.551	0.618	0.062	0.414	0.078	0.012*	0.526	0.319	0.518	0.018*	0.249	0.731
Marital status	Married	0	0.117	0.049	0.019	0.117	0.532	0.125	0	0.004	0.005	0.002	0.02
	Unmarried	0.02	0.318	0.978	0.066	0.091	0.37	0.138	0.838	0.408	0.501	0.182	0.755
	PLS-MGA	0.639	0.771	0.416	0.633	0.376	0.662	0.347	0.103	0.969	0.367	0.701	0.309
Education	≤Bachelor's degree	0	0.163	0.354	0.021	0.644	0.072	0.759	0.001	0.011	0.083	0.019	0.165
	>Bachelor's degree	0	0.012	0.396	0.001	0.527	0.346	0.754	0.074	0.323	0.057	0.092	0.316
	PLS-MGA	0.475	0.138	0.784	0.911	0.779	0.464	0.606	0.409	0.295	0.56	0.324	0.619
Monthly expenditure	≤20 million	0	0.14	0.901	0.066	0.027	0.002	0.119	0.08	0.002	0.949	0.008	0.414
	>20 million	0.02	0.806	0.268	0.318	0.679	0.473	0.028	0.036	0.261	0.002	0.086	0.113
	PLS-MGA	0.185	0.597	0.493	0.719	0.323	0.287	0.403	0.529	0.177	0.018*	0.703	0.408
EV driving experience	≤1 year	0	0.176	0.288	0.007	0.112	0.910	0.021	0	0.12	0.438	0.021	0.151
	>1 year	0.02	0.287	0.749	0.330	0.138	0.014	0.184	0.077	0.363	0.841	0.376	0.026
	PLS-MGA	0.673	0.910	0.805	0.219	0.041*	0.057	0.672	0.442	0.973	0.559	0.882	0.39

electric vehicles on daily activities and the enhancement of work productivity. This is consistent with prior research demonstrating the tangible advantages of EV in routine use (García de Blanes Sebastián et al., 2024; Zhou et al., 2021).

Price value represents the second most influential factor affecting consumers' intention to purchase electric vehicles (EVs) in Indonesia. With a coefficient of 0.355 (p-value <0.05), it demonstrates a relatively strong and statistically significant relationship with purchasing intention. The results suggest that the negative perception of high upfront costs can be alleviated through government incentives and the prospect of lower long-term operating expenses. The analysis indicates that Indonesian consumers prioritize purchase subsidies, maintenance expenses, and affordable vehicle prices when selecting electric vehicles. This is consistent with findings from previous studies, which reported that price value significantly enhances EV purchase intentions among Generation Y consumers in Malaysia (Vafaei-Zadeh et al., 2022). Although electric vehicles are generally considered expensive, studies in India have shown that people perceive the price as an inherent aspect of innovative technology. The higher the perceived price value (benefits outweighing costs), the more likely consumers are to purchase electric vehicles (Asadi et al., 2021; Krishnan & Koshy, 2021; Li et al., 2017). Nonetheless, contrasting evidence from other studies suggests that price value may not always have a significant effect on EV purchase intentions (Gulzari et al., 2022; Manutworakit & Choocharukul, 2022; Xu et al., 2019).

Personal innovativeness is identified as the third factor influencing the intention to adopt EVs in Indonesia. With a coefficient of 0.205 (p-value <0.05), it reflects a moderately strong and statistically significant relationship with purchase intention. This indicates that individuals who exhibit a higher degree of openness to innovation are more likely to explore, evaluate, and adopt emerging technologies such as EVs. Generally, highly innovative individuals embrace creative concepts and products and can influence others to do the same. This finding supports prior studies, even when respondents were aware of electric vehicles but did not own them (He et al., 2018; Khazaei & Tareq, 2021; Shanmugavel & Micheal, 2022). When analyzing the behavior of existing EV users in Indonesia, personal innovativeness appear to have a direct impact on adoption decisions, as reflected by a coefficient value of 0.200 (p-value <0.05). This suggests that for current users are significantly driven by the novelty, features, or technological sophistication of electric vehicles.

Hedonic motivation represents the fourth factor influencing consumers' intention to adopt EVs. With a coefficient value of 0.252 (p-value <0.05), it demonstrates a relatively strong and statistically significant relationship with purchase intention. In the Indonesian context, hedonic motivation reflects consumers' perception of EVs as enjoyable, novel, and emotionally rewarding products. The sense of excitement, pride, and identity associated with driving an EV contributes positively to consumer interest and intention to adopt. This result is consistent with prior studies indicating that hedonic motivation plays a critical role in shaping attitudes toward EV adoption (Gunawan et al., 2022; Singh et al., 2023; Zhou et al., 2021).

In contrast, social influence exhibits a significant negative effect on behavioral intention, as indicated by a coefficient oSf −0.175 (p-value <0.05). This suggests that higher perceived social pressure or expectations from others are associated with a reduced likelihood of adopting EVs. Rather than encouraging adoption, social influence in this context may create skepticism or resistance, potentially due to prevailing norms, misconceptions, or a lack of widespread acceptance of EVs within certain communities. In this context, social influence refers to societal trends, expectations from others, advertisements, social media perceptions, and government programs that may shape an individual's decision to use electric vehicles. The significant negative relationship suggests that when individuals feel substantial social pressure, they may become less inclined to adopt electric vehicles. This could be because social influence fosters negative perceptions or reinforces adherence to traditional norms, such as continuing to use ICE vehicles (Haustein & Jensen, 2018; He et al., 2022). Interestingly, this finding contrasts with the usual trend in which social influence typically encourages the adoption of EVs (Kapser & Abdelrahman, 2020; Khazaei & Tareq, 2021; Sang & Bekhet, 2015).

Effort expectancy does not appear to influence individuals' intention to use electric vehicles in the Indonesian context. The perceived ease of learning, operating, and understanding electric cars does not necessarily translate into a stronger intention to adopt them. This finding contrasts with prior studies conducted in Pakistan and China, where effort expectancy was found to have a significant impact on consumers' intention to purchase electric vehicles (Lee et al., 2021; Zhou et al., 2021).

Among Indonesian electric vehicle consumers, facilitating conditions are not perceived as a significant factor in shaping the intention to adopt electric cars. However, they play a crucial role in influencing actual purchase decisions within the Indonesian context. With a coefficient of 0.18 (p-value <0.05), facilitating conditions show a statistically significant, albeit modest, association with EV purchasing behavior. These conditions encompass various infrastructural and service-related factors, including the availability of 24-hours customer service, ease of transaction processes for buying and selling EVs, public charging infrastructure, device connectivity, and access to technical support and spare parts. Such elements contribute to consumers' perceptions of how well-equipped the environment is to support EV ownership and use, ultimately influencing their readiness to commit to a purchase (Hagman & Stier, 2022). Sales services play a vital role in product matching, facilitating test drives, and ultimately influencing adoption decisions. After-sales service and spare part availability are also critical for EV adoption (Habib et al., 2024). Compared to ICE vehicles, the relatively limited network of EV repair centers can be a source of frustration for owners (Quak et al., 2016). Among the various facilitating conditions, the availability of public charging infrastructure stands out as particularly influential. Access to reliable and conveniently located charging stations helps reduce range anxiety and minimizes charging delays, especially for users traveling long distances or operating outside their home charging range (Anderson et al., 2018; Helveston et al., 2015; Jain et al., 2022).

Similar to facilitating conditions, habit does not significantly influence the initial intention to adopt EVs; however, it plays a noteworthy role in actual usage behavior. In this study, habit demonstrates a coefficient of 0.216 (p-value <0.05), indicating a moderately strong and significant association with EV adoption. For Indonesian consumers, behavioral patterns such as a tendency to frequently change vehicles, collect automobiles, or remain engaged with emerging technologies contribute to EV usage. These findings diverge from prior research, which has generally reported that habitual behavior does not significantly affect the transition from ICE vehicles to EVs (Gunawan et al., 2022; Moons & De Pelsmacker, 2015; Singh et al., 2023; Zhou et al., 2021). This discrepancy may stem from differences in respondent types, as previous studies only included individuals aware of EVs but not actual owners.

In contrast to studies in other countries, this analysis deliberately limits discussion to the Indonesian context to avoid cross-country overgeneralization. The country's unique infrastructural limitations, economic disparities, and policy frameworks necessitate a localized analysis of adoption behavior.

The findings also provide concrete implications for policymaking. The significant role of price value underscores the importance of expanding subsidies, tax incentives, and financing schemes to reduce perceived financial barriers. Similarly, the strong influence of facilitating conditions highlights the urgency of developing reliable, accessible public charging networks to reduce range anxiety. By addressing these structural and economic barriers, policymakers can accelerate adoption and help align public behavior with national decarbonization targets.

6. Conclusion

The proposed model for electric vehicle adoption in this study is UTAUT3, which explains approximately 60.7% of the intention to adopt EVs and 62.5% of adoption behavior in Indonesia. The results indicate that facilitating conditions, habit, personal innovativeness, and behavioral intention, significantly influence EV use behavior. Among these, behavioral intention emerges as the strongest predictor of EV adoption, shaped by several key factors: hedonic motivation, performance expectancy, price value, and personal innovativeness. Conversely, social influence exerts a significant negative effect on behavioral intention, contrary to findings in other cultural contexts. While facilitating conditions and habit do not significantly impact the initial intention to adopt EVs, they play a crucial role in shaping post-adoption behavior. These results suggest that practical considerations and behavioral tendencies become more prominent once consumers transition from intention to actual use. The divergence in findings compared to studies in other countries may be attributed to differences in respondent characteristics, such as the distinction between EV owners and non-owners.

By focusing on actual EV owners, this study contributes context-specific evidence for Indonesia, highlighting the interplay between structural conditions, personal attitudes, and adoption behavior. The findings emphasize that while government targets are ambitious, adoption will depend on practical measures such as expanding subsidies, developing infrastructure, and addressing public perceptions.

This study has several limitations. Indonesia is a diverse country with varied cultural, geographical, social, economic, and infrastructural conditions across its regions. These differences could lead to varying findings if a broader sample of respondents from across the country were included. Additionally, the dominance of certain EV types in specific regions, with varying performance characteristics, could influence consumer perceptions and responses. The number of EV users and owners is expected to grow across Indonesia, allowing for broader studies.

Future research should broaden the respondent base beyond major metropolitan areas to capture Indonesia's regional diversity and explore the perspectives of potential users who have yet to adopt EVs. Such studies can provide more comprehensive insights into the challenges and opportunities of scaling EV adoption in emerging economies.

Ethical statements

This study involved humans as respondents (non-interventional study). No experiments or clinical trials were conducted. We ensured the confidentiality of all participant data collected during the research process.

Availability of data and material

All the data supporting the findings of this study are available upon request.

Declaration of AI technology usage

ChatGPT: Assisted in grammar checking.

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Conflict of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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